Part:1

Computer Network Fundamentals

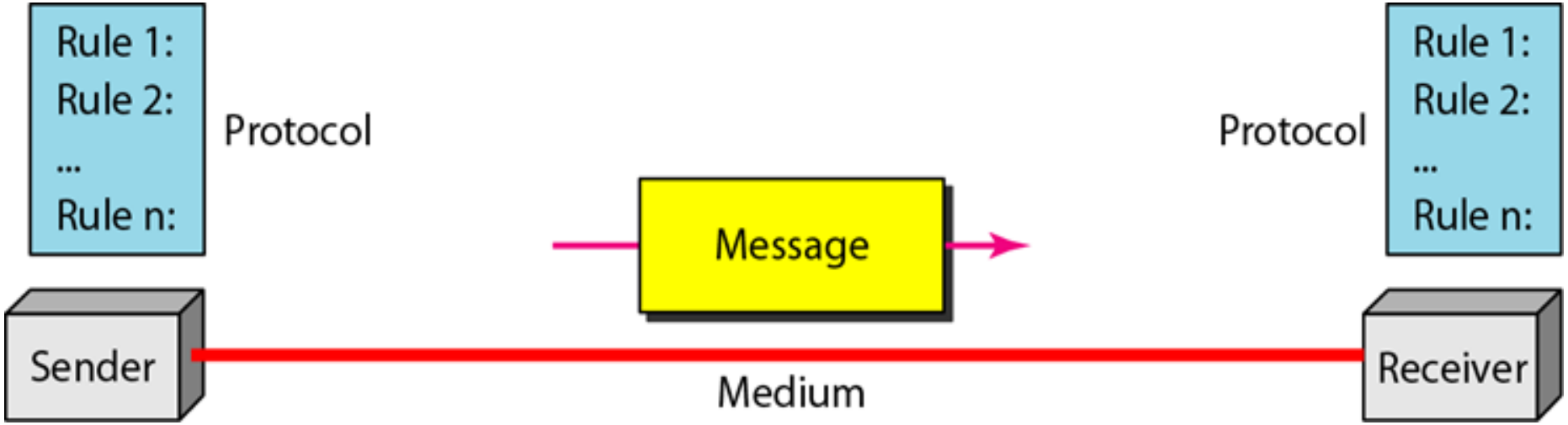
Use of Computer Network

- Simultaneous access to programs and data.
- Sharing peripheral devices like printers, scanners etc.
- Personal communications using email.
- Making backup of information.
- Aiding communication by teleconferencing and videoconferencing.
- Protecting information by account name and password

DATA COMMUNICATIONS

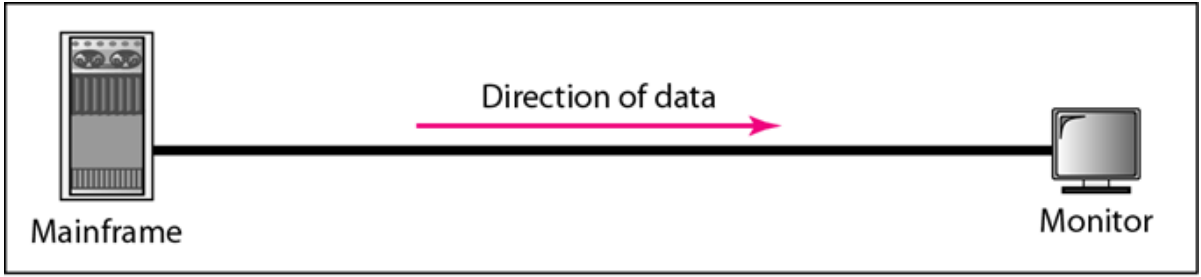
Data communication is the exchange of digital information between two devices or systems. It involves the transmission and reception of data through a medium such as cables, optical fibers, or wireless signals. This process allows computers, smartphones, and other devices to share information, enabling communication, sharing of resources, and access to networks and the internet.

Components of a data communication system

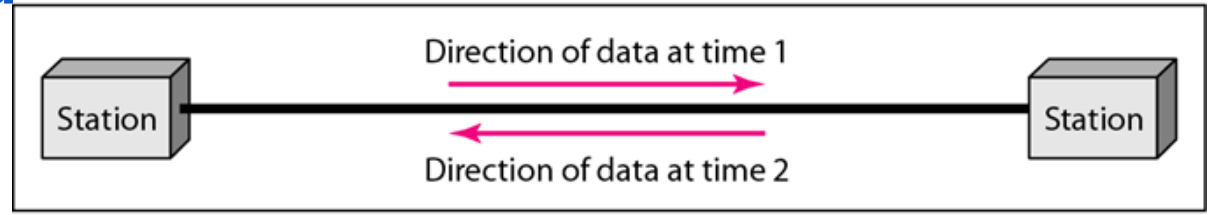


Data flow:

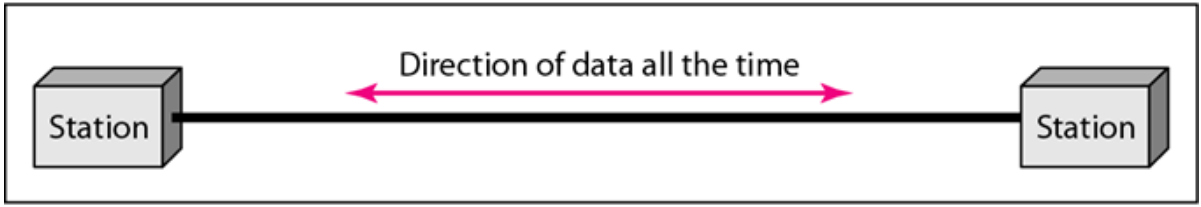
- simplex,
- half-duplex, and
- full-duplex



a. Simplex



b. Half-duplex



c. Full-duplex

Data flow

- **Simplex:**
  - Communication is unidirectional
  - Only one of the two can transmit, the other can only receive
  - Example: Keyboard(input) and traditional monitors(accept output)
- **Half-duplex:**
  - Each station can both transmit and receive, but not at the same time
  - Examples: Walkie-talkies
- **Full-duplex:**
  - Both station can transmit and receive simultaneously
  - Examples: Telephone networks

Computer Networks

A computer network is a system in which multiple computers are connected to each other to share information and resources.

By the simplest definition in the data world, a computer network is a means to connect two or more computers together for the purpose of sharing information.

The computers can be geographically located anywhere.

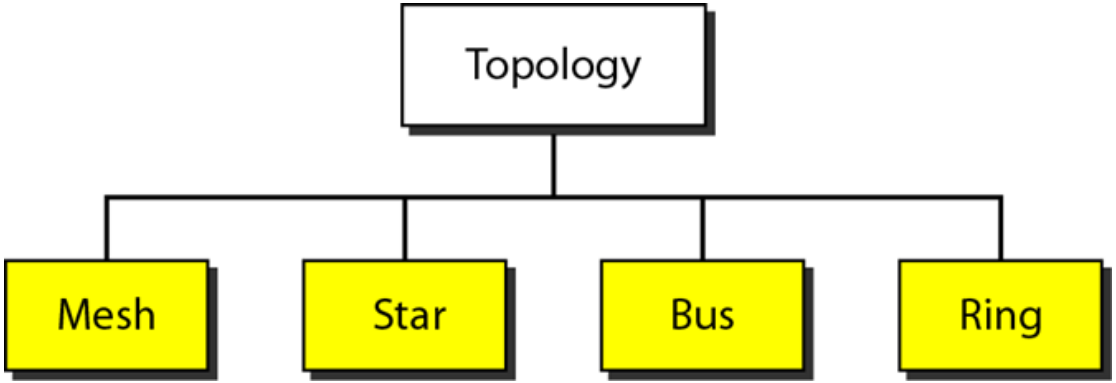
Some network terminologies

- Network topology
- Links
- Media
- Network protocols

Network Topology

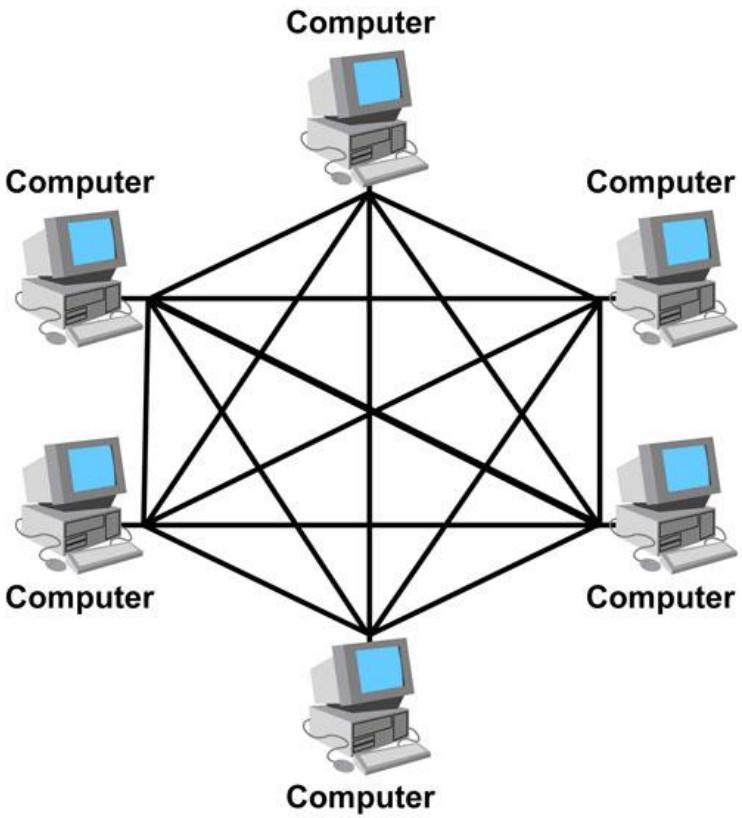
Topology

- The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links from a topology.
- The topology of a network is the geometric representation of the relationship of all the links and linking devices(usually called nodes) to one another.
- There are 4 basic topologies possible:
    - Mesh
    - Star
    - Bus
    - Ring



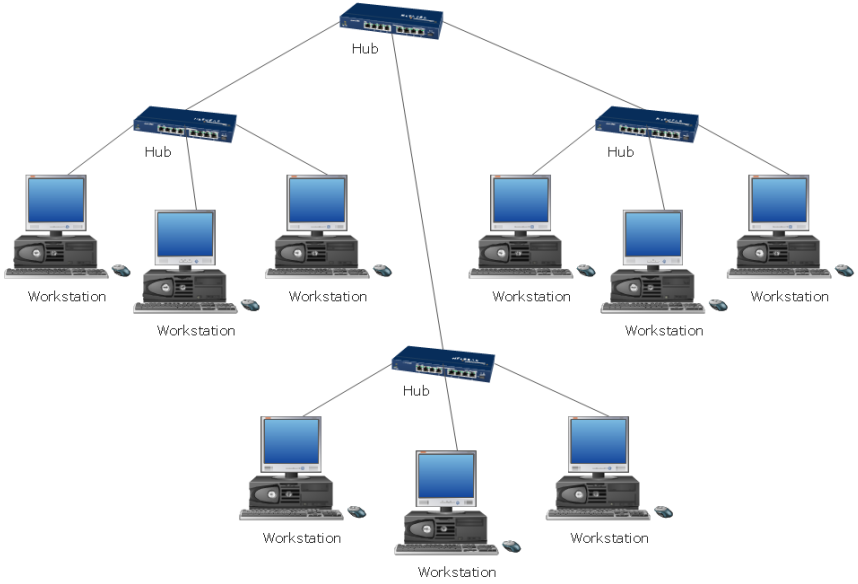
Mesh Topology

- The mesh topology connects all devices (nodes) to each other
- It is used in WANs to interconnect LANs and for mission critical networks like those used by banks and financial institutions.
- Implementing the mesh topology is expensive and difficult.



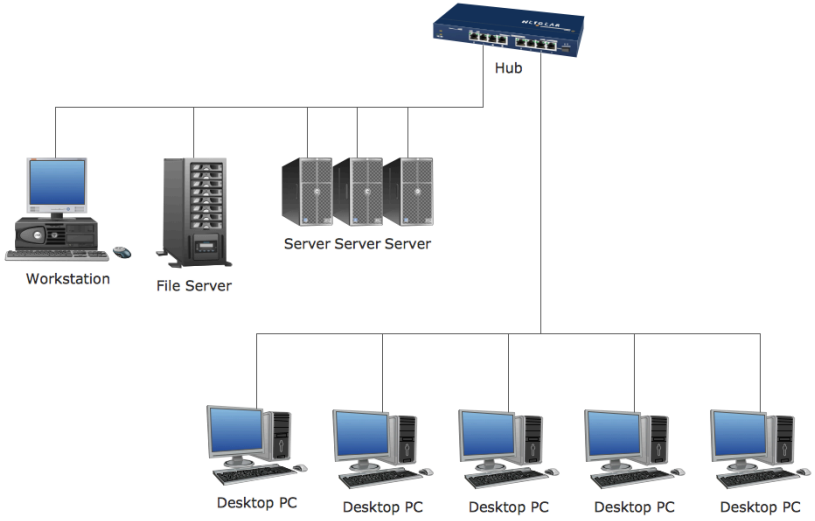
Star Topology

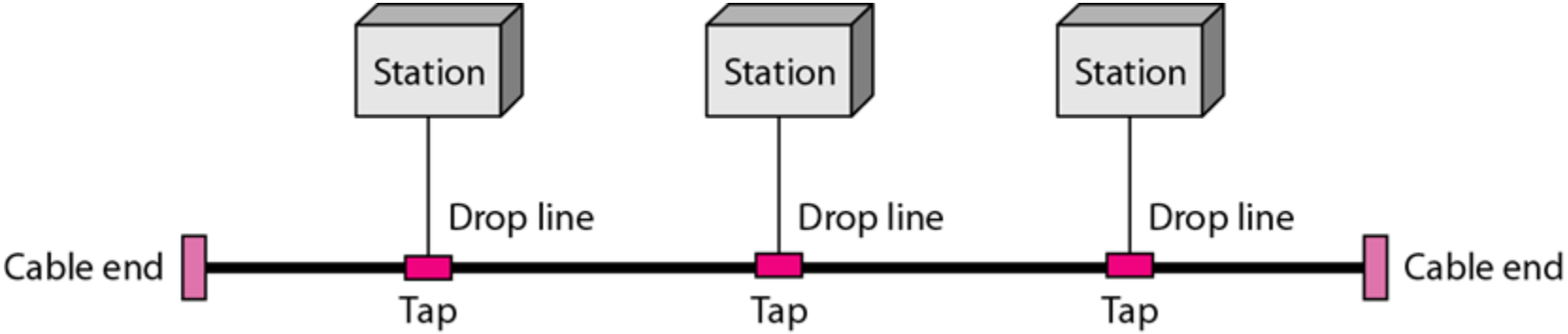
- The star topology is the most commonly used architecture in Ethernet LANs.
- Larger networks use the extended star topology also called tree topology. When used with network devices that filter frames or packets, like bridges, switches, and routers, this topology significantly reduces the traffic on the wires by sending packets only to the wires of the destination host.



Bus Topology

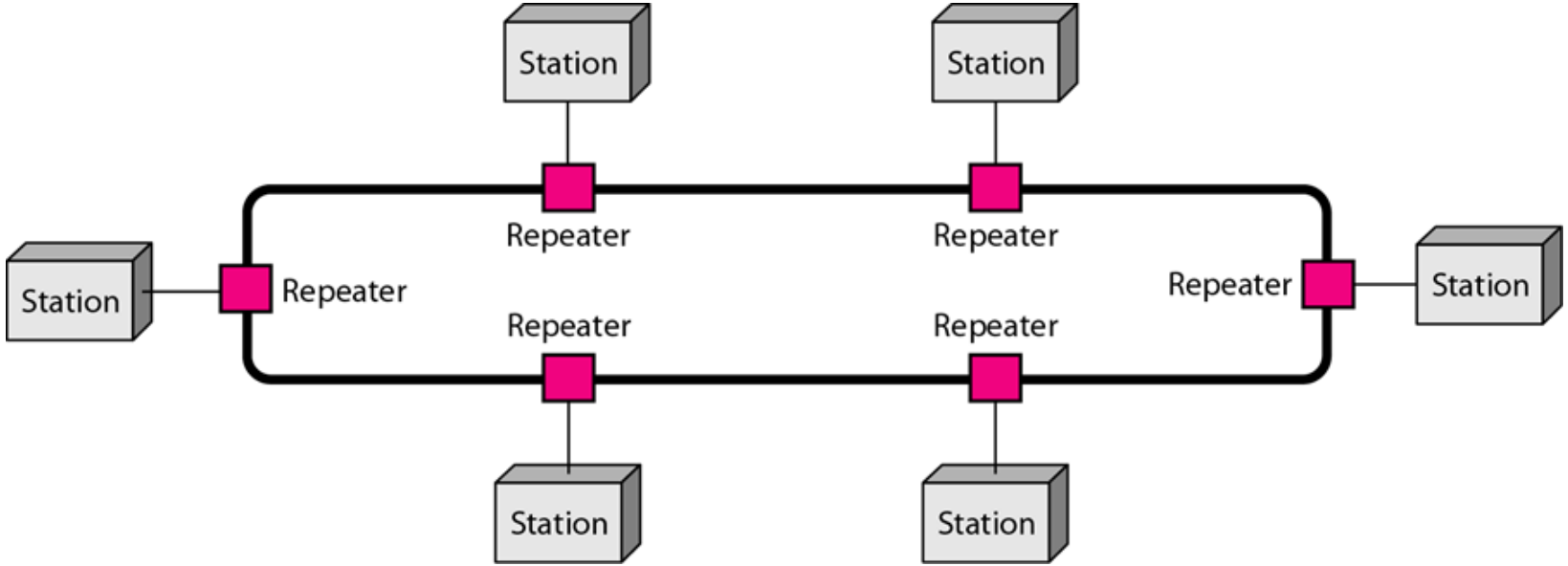
- Commonly referred to as a linear bus, all the devices on a bus topology are connected by one single cable.

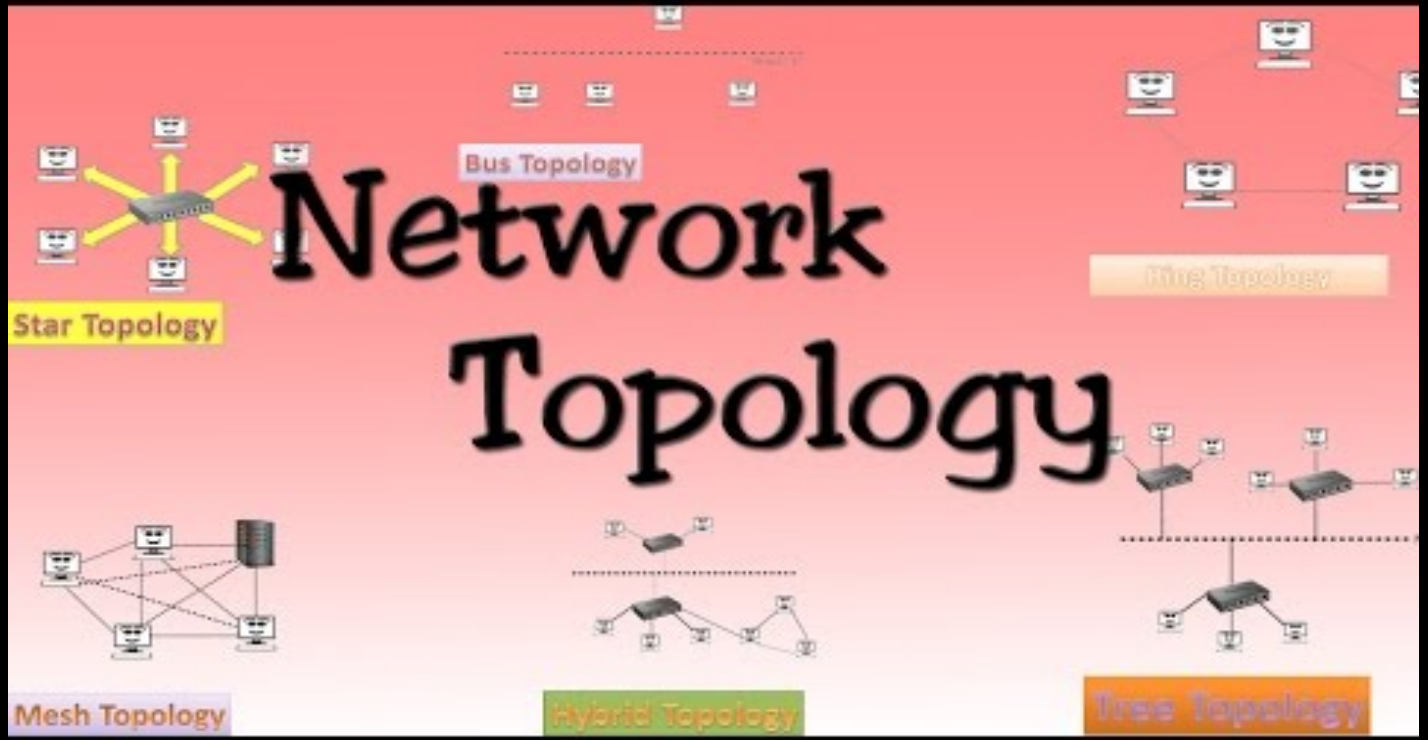




Ring Topology

- Each device has a dedicated point to point connection with only the two devices on either site of it. A signal is passed along the ring in one direction, from device to device until it reach to its destination.




Network
Topology
Bus Topology
Star Topology
Ring Topology
Mesh Topology
Hybrid Topology
Tree Topology

Part: 2

Categories of Network

Categories of Network

Computer network can be classified according to their geographical area and their uses.

- Depending upon the **geographical area** covered by a network, it is classified as:
  - **Local area network(LAN)**
  - **Wide area network(WAN)**
  - **Personal area network(PAN)**
  - **Metropolitan area network(MAN)**
  - **Campus Area Network (CAN)**
  - **Home Area Network(HAN)**
- Depending upon the purpose by a network, it is classified as:
  - **Storage area network(SAN)**
  - **Enterprise private network(EPN)**
  - **Virtual private network(VPN)**

Local Area Network(LAN)

A LAN is a small and privately owned network to provide local connectivity within a small region. "A region” refers to the same office, the same building, the same company and in the same schools.

Characteristics of a LAN include:

- Covers a small geographic area such as a room, office, lab, or building.
- High data transfer speed compared to MAN and WAN.
- Owned, managed, and controlled by a single organization.
- Low installation and maintenance cost.
- Uses wired (Ethernet) or wireless (Wi-Fi) technologies.
- Connects limited number of devices like computers, printers, and servers.

Local Area Network(LAN)

Advantages and Disadvantages of LAN

Advantages

- High speed communication inside the network.
- Low cost installation and maintenance.
- Easy file and resource sharing, including printers and storage.
- Centralized data and software management.
- Improved communication through messaging, VoIP, etc.
- Better security control since the network is locally managed.

Disadvantages

- Limited coverage area cannot connect far distances.
- Security risks if unauthorized users gain physical access.
- Performance drops if many users are connected at the same time.
- Network failures can disrupt the entire LAN.
- Maintenance requires skilled IT personnel, especially for large LANs.

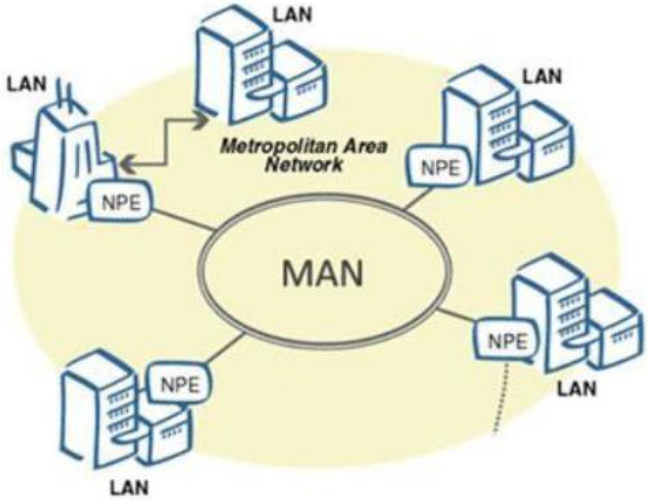
Metropolitan Area Network (MAN)

MAN or Metropolitan area Network covers a larger area than that of a LAN and smaller area as compared to WAN. It connects two or more computers that are apart but resides in the same or different cities. MAN is commonly used by:

- Universities
- Government organizations
- Large companies with multiple regional offices
- Internet Service Providers (ISPs

Characteristics of MAN:

- Covers a city-wide area larger than a LAN but smaller than a WAN.
- High-speed communication using technologies like fiber optics.
- Connects multiple LANs within a metropolitan region.
- Moderate data transmission speed: faster than WAN.
- Often managed by a single organization or service provider.
- Supports shared resources across several buildings or campuses.



Metropolitan Area Network (MAN)

Advantages and Disadvantages of MAN

Advantages

- High-speed data transfer across a large city area.
- Cost-effective compared to WAN because the area is limited.
- Easy to expand by adding more LANs.
- Efficient resource sharing such as files, internet, and applications.
- Improved communication between organizations in the same city.
- Better security than WAN since the coverage area is smaller.

Disadvantages

- Installation cost is high compared to a LAN.
- Maintenance is complex due to large geographical coverage.
- Requires technical expertise to manage and troubleshoot.
- Higher chances of network congestion if many LANs are connected.

Wide Area Network(WAN)

A WAN is used to transmit data and information over large geographical distances which may even span entire countries and continents. A WAN can contain multiple smaller networks, such as LANs or MANs. The Internet is a WAN, and connects computers all around the world together. Communication between networks is called internetworking.

Characteristics of a WAN:

- Covers a very large geographical area — country, continent, or even global.
- Connects multiple LANs and MANs over long distances.
- Uses public or private telecommunication networks (satellites, fiber optics, leased lines).
- Lower data transmission speed compared to LAN and MAN.
- Operated by multiple organizations or ISPs due to the large area.
- Supports long-distance communication for businesses and institutions.
- Highly scalable, can connect new networks across cities/countries.

Wide Area Network(WAN)

Advantages and Disadvantages of WAN

Advantages

- Enables communication across large distances (global connectivity).
- Allows resource and data sharing between branches around the world.
- Supports centralized data management for multinational organizations.
- Highly scalable and easy to add new locations.
- Useful for remote work, cloud services, and distributed systems.
- Multiple communication technologies provide flexibility.

Disadvantages

- Very expensive to set up and maintain due to long-distance connections.
- Higher latency compared to LAN and MAN.
- Lower data transfer speed and more transmission delay.
- Complex infrastructure requiring skilled technicians.

Thank You